

The Modeling of Artificial Neural Network of Early Diagnosis for Malnutrition with Backpropagation Method

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Abstract— The negative effects of malnutrition can be minimized by developing medical technology through combining expert experience and knowledge to produce an early diagnosis. The development of ANN architectural model is conducted to identify the types of malnutrition. This research consisted of 2 phases, which were training phase in which it generated ANN weight by using feed-forward of activation function, and testing phase in which the result of the previous stage was tested to obtain output. The resulting architectural model has a 96% accuracy rate with MSE of 0.000997 during 5 seconds training period. Regression results show that the resulting model has a high degree of accuracy to produce output of malnutrition types such as marasmus, kwashiorkor, and marasmus-kwashiorkor.

Keywords— Early diagnosis, Malnutrition, ANN, Backpropagation.

I. INTRODUCTION

The vision and mission of the Department of Health are in line and consistent with the vision and mission of the President of the Republic of Indonesia which is realizing Indonesia which is sovereign and independent, and has personality based on mutual cooperation. The vision is stated in seven Development Missions which include improving the society's health level by priority agenda of the Working Cabinet or *Nawa Cita*. One of the efforts to improve the health level is improving people's nutrition because nutrition is a process where organism uses food they consume normally, through digestion, absorption, transportation, storage, metabolism and excretion of unused substances to maintain life, growth and normal functions of the organs, as well as producing energy.

In some developing countries, there are still malnutrition cases commonly found. Food shortage and malnutrition are serious problems which are still faced by most Indonesians, especially the poor. Despite general improvement in food availability, health and social services, famine and malnutrition happen in nearly all regencies in Indonesia in different forms. This problem will certainly have an adverse effect on their health in the future, such as disabilities and even death [1]. In most cases, malnutrition is detected when the patient's condition is already severe. Due to its negative effect,

it is necessary to diagnose malnutrition as early as possible in order to make a quick medical action [2].

ANN model with backpropagation method is an artificial intelligence model that has a high accuracy level of training. ANN method has good training capabilities with predictive levels that are sufficient to identify psychological disorders so that the method can help to determine the diagnosis based on the results of the analysis using ANN method. Artificial Intelligence (AI) is a part of one of the computer sciences that studies how to make machines that can work like and as well as those done by humans and even better than what the humans do. Artificial neural network is an effective, accurate, and simple tool to assist doctors in analyzing, modeling and understanding medical data [3]. In addition, ANN can reduce manual diagnostics errors since the system can combine the experience and knowledge from several doctors [4]. Because humans are clever in solving problems because humans have knowledge and experience. Knowledge is gained from learning. The more knowledge they have, the more able the problem to be solved. However, the provision of knowledge alone is not enough, humans are also given the brain to think, draw conclusions based on knowledge from the experience. Without having the ability to think well, people with experience and knowledge will not be able to solve problems properly. If the ability to think is very good, but there is no provision of sufficient knowledge and experience, people will not be able to solve the problem properly.

There are several previous studies stated that ANN technique is widely used because it is proven to have a good error and accuracy rates. The multilayer structure of artificial neural network is the most commonly used due to its very high success rate in diagnosing a disease. Multilayer method used in this study is the backpropagation neural network (BPNN). This method is selected because it can explain the relationship between the existing dataset by imitating the function and works of human brain by using the trial-and-error [5].

Therefore, the research question in this paper is "How to develop an architectural model of BPNN to perform early diagnosis on malnutrition?" It is expected that the developed model can serve as a medical decision support for patient-

specific assesment and advice generic consultation by providing diagnostic and prognostic inferences [6].

II. PREVIOUS RESEARCH

In the developing countries, the use of accessible electronic healthcare services can help to realize a sustainable healthcare, which provide benefits for the patients and healthcare professionals through the better medical decision-making [7]. Malnutrition in developing countries can also be helped through the use of agent technology that can help detecting malnutrition and spreading the knowledge concerning malnutrition to the people [8]. Meanwhile, the use of ANN method to assist the detection process of malnutrition, especially in children, can also provide a fairly high level of accuracy up to 77% [2]. The use of ANN may also provide a low error rate in detecting malnutrition. In addition beside provide good results in detecting malnutrition, the ANN is also widely used to detect various diseases. There are several previous studies which stated that ANN can be applied to the process of diagnosis of the diseases, such as urinary diseases, heart diseases, chest diseases, liver diseases, tuberculosis, as well as providing a high level of accuracy [3] [9] [10] [11]

Artificial Neural Network is an implementation of artificial intelligence technology that adopts the human nervous system so that it works like a human brain system that has the ability to process information including learning, remembering information, equating data and determining patterns during the learning process. In the journal of Disease-Free Survival Assessment by Artificial Neural Networks for Hepatocellular Carcinoma Patients After Radiofrequency Ablation states that the Artificial Neural Network model can recognize data patterns through the learning process and has been applied to obtain the support of the medical decision [12]

The journal titled "An Artificial Neural Network Integrated Pipeline for Biomarker Discovery Using Alzheimer's Disease as a Case Study" explains that Artificial Neural Networks is a form of innovative learning, a statistical model that mimics the function of neurons, able to identify pattern and separate it linearly by giving a numerical weight score to each input and adjusting it to the data sample. [13] In addition, Artificial Neural Networks can use parallel processing to predict solutions to complex and non-fixed data.

In Figure 1, the input vector consists of a number of values (features) that is given as the input values in ANN, the input vector consists of values (i_1, i_2, i_3, i_4, i_n), each input value passes through a contact of w , then all values are combined. This combined value is then processed by the activation function to produce signal z as outputs. The activation function uses a threshold value to limit the output value so that it is always within the specified threshold value.

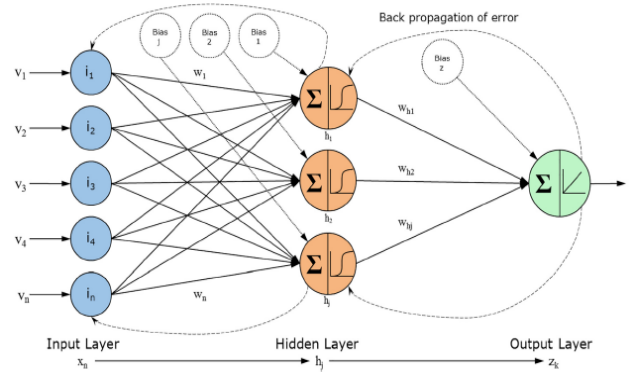


Figure 1. Diagram of the Artificial Neural Networks
Source: D. Zafeiris et al, 2018

The artificial neural networks have an activation function that is used to control the output of neurons so that it matches to the specified output value. Some activation functions that are often used in artificial neural networks.

The second previous research titled Adaptive Gradient Descent Backpropagation for Classification of Breast Tumors in Ultrasound Imaging compares three gradient descent algorithms based on backpropagation method to classify breast tumors to distinguish benign and malignant tumors. The three algorithms are Gradient Descent (GD), Gradient Descent with Momentum (GDM) and Adaptive Gradient Descent (AGD) where AGD algorithm outperforms other algorithms because it has the highest ability to classify benign and malignant tumors with achieving the classification accuracy of 84.6% [14]

In general, the learning algorithms can be categorized into learning algorithms of supervised learning and unsupervised learning. Supervised learning is using input-output vector mapping which is the training data set. Whereas, unsupervised learning is not using the training data sets. The backpropagation algorithm discussed below is a learning algorithm that is supervised learning which is generally for multi-layer artificial neural networks.

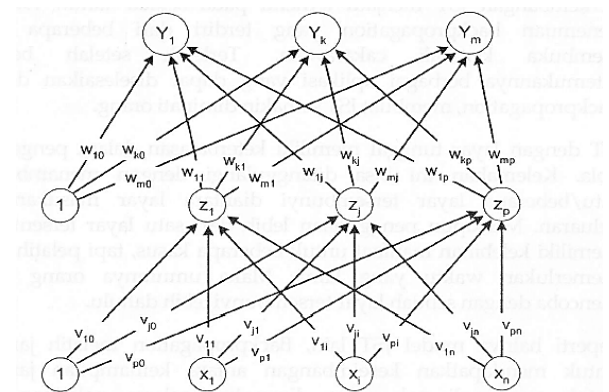


Figure 2. Architecture of Backpropagation
(Source: Jong Jek Siang, 2009)

The multi-layer neural network consists of several layers of artificial neurons and the layer is connected through a

path that is connected to one another. The backpropagation learning algorithm consists of three stages: forward propagation of the input signal, back propagation of the error, and update weight. By spreading the input signal forward (output), the first backpropagation algorithm calculates the output for each vector to be included in the training data set. Then, modify the weight for each input vector. Errors in the output neurons spread to the backward direction from the output layer to the input layer. Multilayer Perceptron (MLP), also called BPNN, is a part of the multilayer forward neural network. BPNN, where this method applies the rule of Widrow - Hoff or Delta rule. BPNN consists of three layers, those are one input layer, one or more hidden layers and one output layer. The factors that can be taken into consideration when designing or building a model architecture on the BPNN are doing architectural initialization, transferring functions in the hidden layer and output layer, training methods and algorithms and determining the parameters used. BPNN has several advantages, such as the resulting architecture that is simple, easy to build the desired model and can produce fast calculation speeds. Some of the disadvantages of the BPNN method are the poor generalization, lack of *program* design that is *rigorous to the theory* and difficult to control the training process. Despite having several obstacles, there are many successful applications of BPNN in various fields. One of the reasons is that users apply experience and apriori knowledge during designing the BPNN.

The purpose of this study is to produce a nonlinear model of the BPNN model in order to be able to perform an initial diagnosis to identify malnutrition in children. The novelty of this study is the determination of input and target with Rules Based. Rules Based is the result of processing the data from symptom data into numerical forms from zero to one. As a result, the data pattern can be processed and known by the Artificial Neural Network. This research was rule based on the determination of inputs and target with novelty of having four different training targets and functions.

It is hoped that this research can contribute in providing malnutrition information and prediction result that can be used by midwives at the integrated health service post and the public in identifying malnutrition in children as early as possible.

III. RESEARCH METHODOLOGY

A. Data Collection

Data was collected in two ways, which are primary data and secondary data. Primary data is data collected directly from sources or in research location, i.e. nutritionist. Secondary data is data collected from books, internet, journal and other literatures related with the research objects. The required data was related to types of malnutrition, symptoms of malnutrition and data of general disorder. The research is quantitative because symptoms of malnutrition was represented by numbers and processed through *Backpropagation* algorithm so the output was calculated numbers consistent with the established rules.

The collected data included:

1. Data of Types of Malnutrition

Table 1. shows types of malnutrition due to lack of carbohydrate or calorie and protein.

Table 1. Data of Types of Malnutrition

No	Type of Malnutrition
1	Marasmus
2	Kwashiorkor
3	Marasmus-Kwashiorkor

2. Data of General Disorder

Table 2. shows general disorder to identify symptoms related with early childhood malnutrition.

Table 2. Data of General Disorders

No	General Disorder of Malnutrition
1	Skin disorder
2	Impaired vision
3	Hair disorder
4	Mental disorder
5	Internal organ disorder
6	Digestive disorder
7	Growth disorder

3. Data of Symptoms of Malnutrition

Table 3 shows symptoms of malnutrition to identify malnutrition in infants.

Table 3 Data of Symptoms of Malnutrition

No	Symptom
1	Face resembling elderly people
2	Dry, cold, loose skin
3	No fat under skin
4	Pink dots, joining into spot which slowly darkens
5	Round and swollen face
6	Clear skeletal muscles
7	Tired (glazed) gaze
8	Occasionally dry and thin hair which falls easily
9	Reddish thin hair similar to corn silk, dull and easy to pull out
10	Uncaring to the surroundings (Apathetic)
11	Crying even after eating due to hunger
12	A lot of crying
13	Reduced breathing

14	Visible ribs
15	Swollen leg (Edema)
16	Muscle shrinkage (Muscle atrophy)
17	Appearance resembling fat child (Sugar Baby) due to swelling (Edema)
18	Concave stomach
19	Frequent defecation (Diarrhea)
20	Reduced appetite
21	Very thin appearance (<i>Vel over been</i>)
22	Non-age appropriate weight (Disrupted growth)

B. BPNN Procedure

The proposed BPNN model development procedures are as follows:

- Step 1 : Preprocess the data for the model, conducted by collecting data using primary data from 14 Community Health Centers located in Nusaniwe and Sirimau Regencies, Ambon, Maluku. The secondary data were obtained by conducting interviews with nutritionists, pediatricians, and parents and through observation conducted in 14 Community Health Centers.
- Step 2 : The making of rule-based based on the expert data. This step was conducted by designing the rule-based using the rules of if-then as existed in the expert system. The making of rule-based was based on the data that have been collected previously.
- Step 3 : The input value was changed to a random value with a range of [0,1]. The input used in the system of data of malnutrition symptoms that have been represented into a numerical form with variables of 0 (No) and 1 (Yes).
- Step 4: Determine the activation function and parameters used in the BPNN model. The activation function was used to limit the output value to match the predefined ones. In the designed model, the activation function used was the binary sigmoid 'logsig'. The iteration will be stopped if the epoch maximum value is set with a variable of 1000, the mean square error of 0.0001 and the learning rate used was with variations of 0.1, 0.25, 0.5, 0.65, 0.75.
- Step 5: Determine the number of hidden layer neurons. The hidden layer neuron was set by using the trial and error, which means that the fastest and most accurate learning result that will determine the number of hidden layer neurons. Variations in the number of hidden layer neurons used were 100, 120, and 75.
- Step 6: Testing parameters of BPNN model. The phase in

which the ANN weights resulted from the training phase will be implemented in the testing phase to generated output. Where the weight of ANN generated from the training phase will be implemented into the testing phase by inputting the symptoms experienced.

- Step 7: The output generated was from the types of malnutrition, among others; marasmus, kwashiorkor, and marasmus-kwashiorkor.

C. The Proposed of BPNN Diagnosis Model

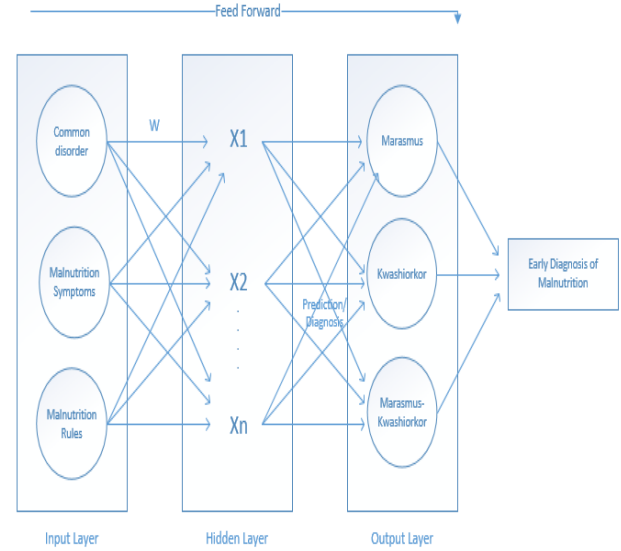


Figure 3. BPNN Diagnosis of Malnutrition Architecture Model

By comparing with the system of work of human brain, the work process of ANN has input called Vectors that consist of 3 values (x_1, x_2, x_3), as each input value passes through a weighted relationship w , then all values are combined. Three values of input are common disorder, malnutrition symptoms, and malnutrition rules. This combination value is then processed by the activation function to generate the y signal as the output. The output are type of malnutrition that can be marasmus, kwashiorkor, or marasmus-kwashiorkor. The activation function uses a threshold value to limit the output value to be always within the specified value limit.

IV. RESULT AND DISCUSSION

This study used 20 training data and 14 testing data. These data were collected based on malnutrition *rules* obtained from nutrition experts.

Table 4. Result of BPNN Architecture Model

No	Learning Rate	Total of Neuron Hidden Layer	MSE	Time	Recognized Data (Data Train + Data Testing)	%
1	0.1	75	0,00105	00:00:05	27	93
2	0.1	100	0,00246	00:00:07	27	93
3	0.1	120	0,00169	00:00:05	27	93
4	0.25	75	0,00148	00:00:04	28	94
5	0.25	100	0,000996	00:00:05	29	95
6	0.25	120	0,000999	00:00:03	27	93
7	0.5	75	0,000999	00:00:02	28	94
8	0.5	100	0,00154	00:00:07	29	95
9	0.5	120	0,000998	00:00:03	28	94
10	0.65	75	0,000998	00:00:02	27	93
11	0.65	100	0,000997	00:00:05	30	96
12	0.65	120	0,000998	00:00:02	27	93
13	0.75	75	0,000998	00:00:02	28	94
14	0.75	100	0,00100	00:00:02	27	93
15	0.75	120	0,000999	00:00:03	27	93

The tables above show the results of the training stage which was performed repeatedly at every point until regression results which approach the targets were obtained. From several artificial neural network architectures which were examined, the most optimal learning rate, total hidden layer neurons to Mean Squared Error, training period and total data correctly recognized by application was point 11. It had the following artificial neural network configuration:

Table 4. Artificial Neural Network Architecture Configuration

Artificial Neutral Network Architecture	Variable Configuration
Maximum Epoch	1000
Error Goal	0,001
Learning Rate	0,65
Total Hidden Layer Neurons	100
Activation Function	Binary Sigmoid (Logsig)

It produced level of accuracy of 96% with Mean Square Error (MSE) of 0,000997 in 00:00:05 training period and it recognized 30 data. Compared with some variations of variable configuration, the most recognized data was 30 data of 34 data (training data and test data). It had low MSE level and fast training period. Below is the result of regression based on the variable configuration above.

Result of model in this paper is attached in figure 3 present the test results of some artificial neural network variables that have been performed based on the ANN network modeling.

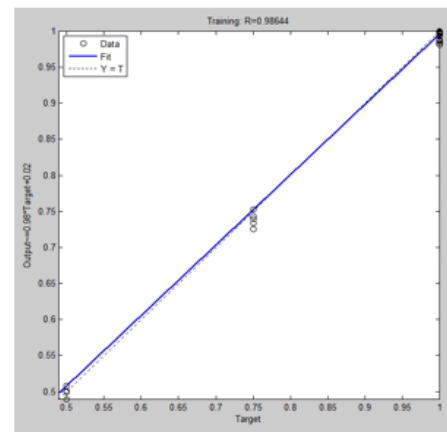


Figure 2. The Result of Regression of Backpropagation Training

The regression result validated network performance, showing whether the output produced by network deviates from target or not. The figure shows that in the result of regression from the training stage, the fit line was almost parallel with the target line most data patterns were near the fit line, indicating that the data inputted into the network can be recognized well by the network.

It resulted in an accuracy of 96% with Mean Square Error (MSE) of 0,000997 in the training process time of 00:00:05 and 30 recognizable data. It can be seen from several variations of variable configurations, the most recognizable data were 30 out of 34 data (training and testing data), as well as having a small MSE level and fairly fast training process time. The regression is performed with Matlab to validate network performance. The output resulted from the network deviates from the target. The regression results obtained from the training phase showed that the fit line was almost parallel to the target line and the data patterns were mostly close to the fit line, which means that the data that would be inputted into the network could be recognized by the network.

The regression result validated network performance, showing whether the output produced by network deviates from target or not. The figure shows that in the result of regression from the training stage, the fit line was almost parallel with the target line most data patterns were near the fit line, indicating that the data inputted into the network can be recognized well by the network.

CONCLUSION

Backpropagation method that was established could facilitate in identifying malnutrition in children with the symptoms that have been determined through interviews with the expert, rule-based, optimal network architecture settings, such as epoch maximum: 1000, goal error: 0.0001, learning rate: 0.65 and the number of hidden layers: 10 neurons, and

the activation functions used were binary sigmoid or logsig functions. The architectural model that was produced reached data testing accuracy of 96%. With this pattern, the identification of malnutrition can be easier to be detected.

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Tabel 1. Data Train

Rules	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12	X13	X14	X15	X16	X17	X18	X19	X20	X21	X22	Target
1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0,5
2	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0,75
3	1	0	0	1	1	0	1	0	0	1	0	0	0	0	1	0	1	0	0	1	0	1	1
4	0	1	0	1	1	0	1	0	1	1	0	1	0	0	1	1	1	0	0	1	0	1	1
5	0	0	1	1	1	0	0	0	1	1	0	1	0	0	1	1	1	0	0	1	0	1	1
6	0	0	0	1	1	1	1	0	1	1	0	1	0	0	1	1	1	0	0	1	0	1	1
7	1	1	1	0	1	1	0	1	0	0	1	0	1	1	0	0	0	1	1	0	1	0	1
8	1	1	1	1	0	1	0	1	0	0	1	0	1	1	0	0	0	1	1	0	1	0	1
9	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0,75
10	1	1	1	0	0	1	1	1	0	0	1	0	1	1	0	0	0	1	1	0	1	0	1
11	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0,5
12	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0,75
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19	1	1	1	0	0	1	0	1	0	1	1	0	1	1	0	0	0	1	1	0	1	0	1
20	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	0	0	0	0	0,5

Tabel 2. Data Testing

Rules	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12	X13	X14	X15	X16	X17	X18	X19	X20	X21	X22	Target
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0,75
2	0	0	0	1	1	0	1	0	1	1	0	1	1	0	1	1	1	0	0	1	0	1	1
3	0	0	0	1	1	0	1	0	1	1	0	0	0	1	1	1	1	0	0	1	0	1	1
4	0	0	0	1	1	0	1	0	1	1	0	1	0	0	1	1	1	1	0	1	0	1	1
5	1	1	1	0	0	1	0	1	0	0	1	0	1	1	0	0	1	1	1	0	1	0	1
6	1	1	1	0	0	1	0	1	0	0	1	0	1	1	1	0	0	1	1	0	1	0	1
7	1	1	1	0	0	1	0	1	0	0	1	0	1	1	0	1	0	1	1	0	1	0	1
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0,5
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14	1	1	1	0	0	1	0	1	0	0	1	0	1	1	0	0	0	1	1	0	1	1	1